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1 (a) Defination: Directional derivative

let $f(x, y, z)$ & $\hat{u}$ is a unit vector

$$\hat{u} = (u_x, u_y, u_z)$$

$$\hat{u} = u_x \hat{i} + u_y \hat{j} + u_z \hat{k}$$

$$\sqrt{u_x^2 + u_y^2 + u_z^2} = 1$$

$$D_{\vec{u}} f(x, y, z) = \lim_{h \to 0} \frac{f(x + u_x h, y + u_y h, z + u_z h) - f(x, y, z)}{h}$$

$$= \frac{\partial f}{\partial x} u_x + \frac{\partial f}{\partial y} u_y + \frac{\partial f}{\partial z} u_z$$

$$= \left(\frac{\partial f}{\partial x} \hat{i} + \frac{\partial f}{\partial y} \hat{j} + \frac{\partial f}{\partial z} \hat{k}\right) \cdot \left(u_x \hat{i} + u_y \hat{j} + u_z \hat{k}\right)$$

$$= \vec{\nabla} f \cdot \hat{u}$$

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$\Rightarrow$ here $\hat{u} \equiv \hat{r} = \frac{x}{r}\hat{i} + \frac{y}{r}\hat{j} + \frac{z}{r}\hat{k}$

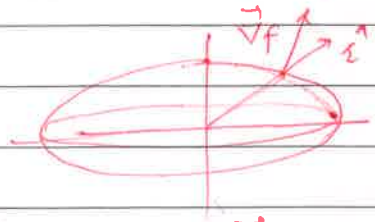
$D_{\hat{r}} f = \nabla f \cdot \hat{r}$

$= \left(\frac{2x}{a^2}\hat{i} + \frac{2y}{b^2}\hat{j} + \frac{2z}{c^2}\hat{k}\right) \cdot \hat{r}$

$$\boxed{D_{\hat{r}} f = \frac{2x^2}{a^2 r} + \frac{2y^2}{b^2 r} + \frac{2z^2}{c^2 r}}$$

(b) $\vec{\nabla} f \cdot \hat{r} = |\vec{\nabla} f||\hat{r}| \cos\theta = |\vec{\nabla} f| \cos\theta$

If $\theta = 0$



when position vector and $\vec{\nabla} f$ is in same direction

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2.

$$\vec{v}_1 = x^2\hat{i} + 3xz^2\hat{j} - 2xz\hat{k}$$

Divergence

$$\vec{\nabla}\cdot\vec{v}_1 = 2x - 2x = 0$$

$$\vec{\nabla}\times\vec{v}_1 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x^2 & 3xz^2 & -2xz \end{vmatrix}$$

$$\hat{i}\left[\frac{\partial}{\partial y}(-2xz) - \frac{\partial}{\partial z}(3xz^2)\right]$$

$$-\hat{j}\left[\frac{\partial}{\partial x}(-2xz) - \frac{\partial}{\partial z}x^2\right]$$

$$+\hat{k}\left[\frac{\partial}{\partial x}(3xz^2) - \frac{\partial}{\partial y}x^2\right]$$

$$= \hat{i}(-6xz) - \hat{j}(-2z) + \hat{k}\,3z^2$$

$$= -6xz\hat{i} + 2z\hat{j} + 3z^2\hat{k}$$

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$$\vec{v}_2 = y^2\hat{i} + (2xy + z^2)\hat{j} + 2yz\hat{k}$$

$$\vec{\nabla}\cdot\vec{v}_2 = 2x + 2y$$

$$\vec{\nabla}\times\vec{v}_2 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ y^2 & 2xy + z^2 & 2yz \end{vmatrix}$$

$$= \hat{i}(2z - 2z) + \hat{j}(0 - 0) + \hat{k}(2y - 2y)$$

$$= 0$$

$$v_3 = \frac{y}{x^2}\hat{i} - \frac{1}{x}\hat{j}$$

$$\vec{\nabla}\cdot v_3 = -\frac{2y}{x^3} - 0 = -\frac{2y}{x^3}$$

$$\vec{\nabla}\times\vec{v}_3 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ \frac{y}{x^2} & -\frac{1}{x} & 0 \end{vmatrix}$$

$$= \hat{i}\times 0 - \hat{j}\times 0 + \hat{k}\left(\frac{1}{x^2} - \frac{1}{x^2}\right) = 0$$

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for Irrotational vector field

$$\vec{\nabla} \times \vec{V} = 0$$

$$\Rightarrow \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ xy & 2x + g(z) & 3yz^2 \end{vmatrix}$$

$$= \hat{i}\left(3z^2 - \frac{\partial g}{\partial z}\right) - \hat{j}\left(0 - 0\right) + \hat{k}\left(2 - 2\right)$$

$$= 0$$

$$\Rightarrow 3z^2 - \frac{\partial g}{\partial z} = 0$$

$$\Rightarrow \frac{\partial g}{\partial z} = 3z^2 \Rightarrow g = z^3 + c$$

Hans